ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2023

B.TECH. CSE ENGINEERING â€" SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) Describe OS functions and structures (Monolithic, Microkernel, Layered architectures). [7 Marks]
(b) Explain Process Control Block (PCB) and process state transition diagram with preemptive vs non-preemptive states. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) What is a System Call? Describe the life cycle of fork(), exec(), wait(), and exit(). [7 Marks]
(b) Differentiate between User-level threads and Kernel-level threads with multi-threading models. [7 Marks]

--- UNIT II ---

Q.2 (a) Given 5 processes with Arrival Times and Burst Times, calculate average turnaround and waiting time using Round Robin (Quantum=2) and SJF. [7 Marks]
(b) Define Critical Section Problem. State the three requirements (Mutual Exclusion, Progress, Bounded Waiting) for a valid solution. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Explain Peterson's Solution for 2-process synchronization. Prove all three requirements are met. [7 Marks]
(b) What are Semaphores? Solve the classical Producer-Consumer bounded buffer problem using counting and binary semaphores. [7 Marks]

--- UNIT III ---

Q.3 (a) Define Deadlock. What are the four Coffman necessary conditions for deadlock occurrence? [7 Marks]
(b) Illustrate Banker's Algorithm for deadlock avoidance with Safety Algorithm using Allocation, Max, and Available matrices. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Describe methods for Deadlock Detection and Recovery in modern operating systems. [7 Marks]
(b) Explain Resource Allocation Graph (RAG) and state how cycles relate to deadlocks in single vs multi-instance resources. [7 Marks]

--- UNIT IV ---

Q.4 (a) Explain Paging memory management. How does Translation Lookaside Buffer (TLB) improve effective memory access time? [7 Marks]
(b) Given reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 with 3 frames, calculate page faults using FIFO, LRU, and Optimal algorithms. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) What is Belady's Anomaly? Demonstrate Belady's Anomaly with FIFO page replacement. [7 Marks]
(b) Explain Segmentation and Segment Table with address translation hardware. [7 Marks]

--- UNIT V ---

Q.5 (a) Compare Disk Scheduling algorithms: FCFS, SSTF, SCAN, C-SCAN, and LOOK with an example seek sequence. [7 Marks]
(b) Explain File Allocation methods: Contiguous, Linked, and Indexed allocation with merits and demerits. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Describe RAID levels (RAID 0, RAID 1, RAID 5, RAID 6) in terms of redundancy and performance. [7 Marks]
(b) Write short notes on: (i) Thrashing and Working Set Model (ii) Access Control Lists (ACL). [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****